

Pre-Requisites

To effectively participate in this session, learners are required to:

- Basic understanding of supervised ML (classification problems).
- Familiarity with logistic regression and/or decision trees at a conceptual level.
- Awareness of the structure of a neural network (layers, weights, activations).
- No coding required, but comfort with following workflows and conceptual math.

Session Summary

This 2-hour live session focuses on understanding the differences between **classical machine learning models** and **deep learning models** through a comparative case study. Learners will first explore how a problem can be solved using a classical ML approach such as *Logistic Regression or Decision Trees*, followed by how the same problem can be approached with a *Multi-Layer Perceptron (MLP)*. The instructor will guide learners step by step, highlighting the contrasts in pipeline, feature handling, data requirements, interpretability, and business outcomes.

Session-related slides and supporting materials for the instructor can be found in the [this](#) folder.

Note: This is a concept-driven session. Learners are encouraged to actively participate in discussions, share their own domain examples, and reflect on situations where classical ML or deep learning would be the more appropriate choice.

Learning Objectives/Outcomes

By the end of this session, learners will be able to:

- Explain the conceptual differences between classical ML models and deep learning models.
- Describe how the same problem can be solved using Logistic Regression / Decision Trees versus a Multi-Layer Perceptron (MLP).
- Compare differences in preprocessing, feature representation, model complexity, and data requirements.
- Interpret the strengths and weaknesses of each approach in real-world scenarios.
- Identify when to prefer classical ML methods and when to apply deep learning models.

Brief Agenda

Table 1: Brief Agenda

Learning Objective	Comments	Duration	Mode
Part I: Introduction	Overview of session goals, importance of comparing classical ML with DL, and outline of session flow.	10 mins	–
Part II: Case Study 1 (Classical ML)	Walkthrough of the first implementation using Logistic Regression / Decision Trees. Show pipeline and outcomes.	25 mins	Presentation

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Table 1: Brief Agenda (Continued)

Learning Objective	Comments	Duration	Mode
Part III: Case Study 2 (Deep Learning)	Demonstration of the same use case solved with a simple Multi-Layer Perceptron (MLP). Highlight differences in pipeline and results.	25 mins	Presentation
Part IV: Comparative Insights	Discuss similarities and differences between classical ML and DL: preprocessing, interpretability, feature handling, and data requirements.	20 mins	Discussion
Part V: Learner Discussion	Learners brainstorm and share contexts where they would apply classical ML vs DL. Encourage examples from their domains.	15 mins	Interactive
Part VI: Q&A	Open floor for learner questions and clarifications on workflows and trade-offs.	15 mins	Open Q&A
Part VII: Conclusion & What's Ahead	Summarize session learnings and preview advanced DL methods (CNNs, RNNs, Transformers).	10 mins	–

Conclusions

By the end of this live session, learners will have strengthened their understanding of the key differences between classical machine learning models and deep learning models. They will be able to map a single use case to both approaches, compare the workflows of Logistic Regression / Decision Trees versus a Multi-Layer Perceptron (MLP), and evaluate trade-offs in data requirements, interpretability, and performance. Learners will also feel more confident in deciding when to use classical ML methods and when deep learning is the more suitable choice for real-world applications.

Additional Readings

The following additional links have been provided as references for additional learning:

- [Machine Learning vs Deep Learning: Key Differences, Models, and Use Cases](#)
- [AI Vs Machine Learning Vs Deep Learning: Understanding The Differences](#)
- [Machine Learning vs Deep Learning: When to Use What?](#)